

Reading

Matching headings

- 1 You are going to read a typical IELTS Reading passage. Before you read the whole passage, look at the title, subheading and first paragraph. What do you think the rest of the passage will be about? Choose one or more from the list below.

- 1 a description of farming techniques in China ☐
- 2 what happens when farms don't get enough rain ☐
- 3 whether or not cloud seeding is a good way of creating rain ☐
- 4 research to test how effective cloud seeding is ☐
- 5 a general description of how cloud seeding is being carried out in one or more places ☐
- 6 a detailed scientific description of how cloud seeding works ☐
- 7 the possible advantages and disadvantages of cloud seeding ☐

The rain makers

Science and technology work with nature to bring rain when and where it is needed

A Wheat farmer Gang Liu is a worried man. The annual rains have not arrived, and there is a danger that unless there is substantial rainfall soon, his annual wheat crop will fail. As he looks anxiously at the clouds which promise rain but are failing to deliver it, there is a sudden loud roar, and from fields for miles around, hundreds of small rockets are fired into the clouds. Within twenty minutes, the farms around the eastern Chinese city of Luohe are experiencing their first rain for many weeks. Gang Liu's valuable wheat has been saved, thanks to a technique known as 'cloud seeding', in which the chemical silver iodide (AgI) is introduced into clouds. This causes the tiny drops of moisture in the clouds to turn to ice. These tiny ice particles join until they become heavy enough to fall from the sky, turning into rain as they melt.

B But did cloud seeding really cause the rain in Luohe to fall, or was it just a coincidence? Experts often question whether cloud seeding actually works. It is hard to tell how effective cloud seeding actually is, they say, as it might have rained anyway, without human intervention. But this has not stopped many governments and organisations from trying. There are currently 150 weather-modifying projects taking place in more than 40 countries. Not all of them are aimed at creating rain. The Eastland

C Scientific Enterprises Corporation in the USA, for example, is experimenting with firing microwaves into clouds to prevent the tornadoes which cause enormous damage to the country every year. In Russia, experiments have been carried out to make sure the sun shines during important national events.

D In China, where the majority of cloud-seeding operations take place, weather-modification authorities use army rockets to fire silver-iodide particles into the clouds. 39,000 staff working for the China Meteorological Administration (CMA) are equipped with 7,113 army cannons which, in 2006, were used to fire a million silver-iodide rockets into the atmosphere. This costs over \$100 million a year, although the CMA claims the results are worth the expense. Between 1999 and 2006, they say, cloud seeding produced 250 billion metric tonnes of rain and prevented thousands of farmers from losing their crops.

E "We want to understand what makes clouds rain," says Philip Brown of the UK Meteorological office, explaining why so much time, effort and money are being invested. "But there is a more powerful economic reason. A lot of countries around the world are at risk from drought, and governments will try anything to make sure that doesn't happen, even if the scientific evidence is weak. The potential economic value is greater than the scientific value. Making it rain might allow you to keep agriculture going where, without human intervention, it might fail."

F Some people are concerned, however, that altering the weather can have negative consequences. Leonard Barrie, director of the research department at the World Meteorological Organisation in Geneva, explains why. "All areas of weather modification are still very controversial. Some people think that diverting water for irrigation benefits some people, but is a disadvantage to others. Someone in one area will get more water, but as a result, someone somewhere else could get less." His fears may be justified. Recently, the town of Zhoukou in China's Henan province accused neighbouring town Pingdingshan of 'stealing' rain from clouds that were due to pass over its own farms, prompting what may be the world's very first documented incident of 'rain rage'.

- 2 Now check your answers to Exercise 1 by quickly reading the rest of the passage.

- ③ Now look at the Reading task below. Before you do the task, use a dictionary to check the meanings of the underlined words in i-ix.

Questions 1-6

The reading passage has six paragraphs, A-F.

Choose the correct heading for each paragraph from the list of headings below.

List of headings

- i Making peaceful use of a military weapon
- ii How modifying the weather has changed the world
- iii What is prompting this research?
- iv A period of drought comes to an end
- v An old solution to a new problem
- vi Winners and losers
- vii Tests provide encouraging results
- viii A waste of money
- ix Global attempts to change the weather

- | | |
|---------------------|---------------------|
| 1 Paragraph A | 4 Paragraph D |
| 2 Paragraph B | 5 Paragraph E |
| 3 Paragraph C | 6 Paragraph F |

- ④ Now read each paragraph carefully and match it to the correct heading.

Sentence completion

- ⑤ Read the task and Questions 7-12 below. Then answer these questions.

- How many words can you write in each answer?
- How many numbers can you write in each answer?
- Can you write a word *and* a number in each answer?
- Look at the sentences and decide which of these types of word could not be used to complete any of the sentences. Tick (✓) one or more word types.

a noun	☐	a verb	☐
an adjective	☐	an adverb	☐

Questions 7-12

Complete the sentences below.

Choose **NO MORE THAN ONE WORD AND/OR A NUMBER** from the passage for each answer.

- Experts are unsure if cloud seeding is or not.
- At the moment, there are over where projects are being carried out to modify the weather.
- Thanks to modern, it is now possible to get better results when clouds are monitored.
- The Desert Research Institute project will finish in a couple of
- The CMA gets the equipment they need from the
- A large number of benefited from cloud seeding carried out by the CMA.

- ⑥ Now do Questions 7-12.

Pick from a list

- ⑦ Look at Questions 13-14 below and underline the key words. Use these to find the right places in the passage.

Questions 13-14

Choose **TWO** letters, A-E.

Which **TWO** of these sentences are true about cloud seeding, according to the passage?

- A China carries out more cloud seeding than anywhere else.
- B Cloud seeding is too expensive for most countries.
- C Cloud seeding is mostly done for scientific rather than economic reasons.
- D Cloud seeding helps turn dry areas of land into agricultural areas.
- E Cloud seeding may affect the distribution of rainfall.

- ⑧ Now answer Questions 13-14.